QUIZ 8 / NOV 03 / RECITATION

Calc III.

Know the moves.

Lessons 23–25: spherical coordinates, mass and center of mass, vector fields. A complete short guide with worked examples and answer key.

01 —

Three lessons. *One quiz.*

Quiz 8 is Tuesday, November 3, in recitation. Purdue's Fall 2026 calendar names Lessons 23, 24, and 25. Those are sections 16.5 (spherical coordinates), 16.6 (mass calculations), and 17.1 (vector fields).

23 / SPHERICAL COORDINATES

Choose coordinates that match a sphere or cone. Describe the solid with bounds on rho, phi, and theta. Multiply by the spherical volume factor before integrating.

24 / MASS AND CENTER OF MASS

Density weights each small piece. Total mass is the integral of density. Center of mass is weighted average position: coordinate moment divided by mass.

25 / VECTOR FIELDS

A field assigns an arrow to every point. Read arrow direction from components, sketch a few test arrows, and recognize when a field is a gradient.

> **ONE-SITTING PLAN**
>
> 10 min: read pages 3–5. 20 min: solve page 6 without notes. 10 min: check page 7; redo misses. Spend most time on bounds and density-weighted integrals.

SOURCE / COURSE SCOPE

Purdue MA 261 Fall 2026 calendar: math.purdue.edu/academic/courses/semester/202710/ma26100/Schedule261.pages.pdf. Topic details cross-checked against the Purdue-hosted MA 261 Exam 2 study guide.

02 —

Spherical coordinates. *Shape first.*

Imagine locating a point by distance from origin, angle around vertical axis, and tilt down from positive z-axis. Call these rho, theta, and phi. Purdue uses phi for tilt, with 0 at north pole and pi at south pole.

`x = rho sin(phi) cos(theta)` `y = rho sin(phi) sin(theta)`

`z = rho cos(phi)` `x^2 + y^2 + z^2 = rho^2`

Bounds: rho at least 0, theta usually 0 to 2pi for full rotation, phi 0 to pi for entire sphere. Sphere radius R means rho = R. Upper half-space z at least 0 means phi at most pi/2. Cone z = sqrt(x^2+y^2) means phi = pi/4.

> WHY THE VOLUME FACTOR APPEARS
>
> A tiny radial move has length d(rho). A tiny tilt sweeps arc rho d(phi). A tiny turn sweeps arc rho sin(phi) d(theta). Multiply lengths: dV = rho^2 sin(phi) d(rho) d(phi) d(theta). Never omit it.

WORKED / VOLUME INSIDE SPHERE, ABOVE CONE

Solid: rho at most 2, above z = sqrt(x^2+y^2). “Above” cone gives 0 at most phi at most pi/4. Full rotation gives 0 at most theta at most 2pi.

`V = int[theta 0..2pi] int[phi 0..pi/4] int[rho 0..2]`

`rho^2 sin(phi) d(rho) d(phi) d(theta)`

`= (2pi)(1 - sqrt(2)/2)(8/3)`

`= (16pi/3)(1 - sqrt(2)/2).`

Check: positive, smaller than upper hemisphere volume 16pi/3. Setting up correct limits is usually harder than integrating.

03 —

Mass and balance. *Weight the pieces.*

Ordinary geometric center treats all area alike. Density delta(x,y) tells how heavy each unit of area is. Mass is sum of density times tiny area. Center of mass averages each coordinate, with the same weight.

Plate: m = double-int[D] delta dA

xbar = double-int[D] x delta dA / m

ybar = double-int[D] y delta dA / m

Traditional moments: M_y = integral of x delta, so xbar = M_y/m. M_x = integral of y delta, so ybar = M_x/m. Subscript names axis of rotation, not coordinate being averaged.

WORKED / NONUNIFORM RECTANGLE

Plate: 0 at most x at most 2, 0 at most y at most 1. Density delta = 1+x: right edge is heavier.

m = int[x 0..2] int[y 0..1] (1+x) dy dx = 4.

M_y = int[x 0..2] int[y 0..1] x(1+x) dy dx = 14/3.

M_x = int[x 0..2] int[y 0..1] y(1+x) dy dx = 2.

(xbar, ybar) = (7/6, 1/2).

Sanity check: xbar = 7/6 is right of geometric center x = 1; ybar stays 1/2 because density does not vary with y.

> SAME IDEA IN THREE DIMENSIONS
>
> For a solid, replace dA with dV. Then m = triple-int[E] delta dV, xbar = triple-int[E] x delta dV / m, and likewise for ybar and zbar. If using spherical coordinates, dV still includes rho^2 sin(phi).

04 —

Vector fields.*Read the arrows.*

A vector field F(x,y) = (P(x,y), Q(x,y)) gives an arrow at every point. First component is horizontal; second is vertical. Plot arrows at (1,0), (0,1), (-1,0), and (0,-1). Their pattern reveals the field.

THREE PATTERNS TO KNOW

`F = (x,y)` points outward; length grows from origin.

`F = (-y,x)` circulates counterclockwise.

`F = (y,-x)` circulates clockwise.

A gradient field comes from a scalar potential f: F = grad(f) = (f_x, f_y). Its arrows point in direction of fastest increase of f. For example, f = x^2+y^2 has grad(f) = (2x,2y), outward.

WORKED / FIND A POTENTIAL

Given F = (2xy+1, x^2+2y), integrate first component with respect to x. Treat y as constant:

`f = x^2 y + x + g(y).`

Differentiate with respect to y: f_y = x^2 + g'(y). Match second component x^2+2y, so g'(y)=2y and g(y)=y^2+C.

`Potential: f = x^2 y + x + y^2 + C.`

> ### COMMON TRAPS
>
> Phi is measured from positive z-axis, not from xy-plane. Spherical dV needs rho^2 sin(phi). For center of mass, divide every moment by mass. For vector fields, test actual arrows; “rotating” is easy to reverse by eye.

05 —

Practice.*Then check.*

Do 1–8 without notes. For integrals, write bounds and the correct measure before computing.

SPHERICAL

1. Give spherical rho and phi for (0,0,-3). What can theta be?

2. Set up and evaluate volume of upper hemisphere x^2+y^2+z^2 at most 9, z at least 0.

3. Set up volume inside rho = 4 and above z = sqrt(x^2+y^2). Evaluate if time permits.

4. Evaluate integral of z^2 over ball of radius 2 using spherical coordinates.

MASS

5. Find mass and center of mass of 0 at most x at most 2, 0 at most y at most 1, with density 1+x.

6. Uniform plate occupies triangle 0 at most x at most 1, 0 at most y at most x. Find mass and center of mass.

VECTOR FIELDS

7. Does F = (y,-x) rotate clockwise or counterclockwise? Give arrow at (1,0).

8. Find gradient of f = x^2 y + 3y. Is F = (2xy, x^2+3) a gradient field?

Quiz-speed check: 1–4 in 12 minutes, 5–6 in 10 minutes, 7–8 in 3 minutes. If time runs out, prioritize setting up each integral correctly.

06 —

Answer key. *Fix the miss.*

1 / SOUTH POLE

rho = 3, phi = pi. Theta is arbitrary: all rotations identify same point on z-axis.

2 / UPPER HEMISPHERE

int[theta 0..2pi] int[phi 0..pi/2] int[rho 0..3]

rho^2 sin(phi) d(rho) d(phi) d(theta) = 18pi.

3 / SPHERE ABOVE CONE

int[theta 0..2pi] int[phi 0..pi/4] int[rho 0..4]

rho^2 sin(phi) d(rho) d(phi) d(theta)

= (128pi/3)(1 - sqrt(2)/2).

4 / Z-SQUARED OVER BALL

z = rho cos(phi), so integrand times volume factor is rho^4 cos^2(phi) sin(phi).

int[theta 0..2pi] int[phi 0..pi] int[rho 0..2]

rho^4 cos^2(phi) sin(phi) d(rho) d(phi) d(theta)

= (2pi)(2/3)(32/5) = 128pi/15.

5 / HEAVIER ON RIGHT

m = 4, xbar = 7/6, ybar = 1/2. Rightward shift is sanity check.

6 / TRIANGLE

m = int[x 0..1] int[y 0..x] 1 dy dx = 1/2.

xbar = (1/m) int[x 0..1] x*x dx = 2/3.

ybar = (1/m) int[x 0..1] (x^2/2) dx = 1/3.

7 / CLOCKWISE

At (1,0), F=(0,-1): arrow points down. On right side of circle, down means clockwise.

8 / GRADIENT

grad(f) = (2xy, x^2+3). Yes: given F = grad(f).